

# Shashank Yadav Kalki

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## PROFESSIONAL SUMMARY

Machine Learning postgraduate with strong foundations in AI, statistical modelling, and dynamical systems. Experienced in building end-to-end ML pipelines and analysing model behavior, reliability, and failure modes. Interested in applying machine learning and data-driven solutions in real-world systems..

## TECHNICAL SKILLS

**Machine Learning:** Regression, Classification, Clustering, Dimensionality Reduction, Time Series Analysis

**Deep Learning:** Neural Networks, CNNs, RNN/LSTM, Model Optimization

**Generative AI (Foundational):** Embeddings, Prompt Engineering, Vector Search, LLM-based Retrieval (RAG concepts)

**Programming:** Python (NumPy, pandas, scikit-learn), R

**Data Engineering:** Data Cleaning, Feature Engineering, Validation, Bias & Error Analysis

**Visualization:** Matplotlib, Seaborn

**Tools & Platforms:** SQL, Git/GitHub, FastAPI, Streamlit

## PROJECTS

### Confabulation Dynamics in Reservoir Computing – Research Project

May 2025 – Sept 2025

*University College Cork | Under Dr. Andrew Flynn*

- Studied how reservoir computing systems generate “confabulations” (plausible but incorrect outputs).
- Analyzed dynamical systems behavior and emergence of untrained attractors.
- Investigated model reliability and failure modes in recurrent neural networks.
- Connected concepts from nonlinear dynamics with machine learning system behavior.

### Heart Disease Prediction – End-to-End ML Pipeline

Jan 2025 – Mar 2025

*Python | Scikit-learn | FastAPI*

- Designed and implemented a complete ML pipeline covering preprocessing, feature selection, model training, and evaluation.
- Compared multiple classification models using accuracy, precision, recall, and ROC-AUC metrics.
- Exposed model predictions via a FastAPI service and built a visualization layer for interpretability.
- Emphasized validation, reliability, and real-world usability of predictive models.

### Human Activity Recognition – Time Series Machine Learning

Oct 2024 – Jan 2025

*Python | PCA | Time Series Analysis*

- Performed time-series feature engineering on wearable sensor datasets.
- Applied dimensionality reduction techniques to improve model efficiency and generalization.
- Evaluated classification models and analyzed error patterns and bias.
- Documented results with focus on data quality and robustness.

### Financial Time-Series Analysis & Predictive Modelling

Aug 2025 – Sept 2025

*Python | Statistical Modelling | Time Series*

- Analyzed historical financial time-series data to identify trends, volatility, and seasonality.
- Applied regression and forecasting techniques to study predictive behavior of indicators.
- Validated datasets and documented modelling assumptions and limitations.
- Produced analytical summaries to support data-driven insights.

### Conversational Analytics Platform (Prototype)

Sept 2024 – Nov 2024

*Python | NLP | FastAPI | Vector Databases*

- Built a prototype conversational system using embeddings for semantic search over document collections.
- Implemented document ingestion, embedding generation, and similarity-based retrieval workflows.
- Explored prompt design and response grounding concepts for reliable LLM outputs.
- Focused on system reliability, scalability, and relevance of responses.

## EDUCATION

### University College Cork

Cork, Ireland

M.Sc. Mathematical Modelling & Machine Learning (Second Class Honours Grade I )

2024 Sept – 2026 March

### Mahatma Gandhi Institute of Technology

Hyderabad, India

B.Tech Computer Science (Data Science)

2020 – 2024

## CERTIFICATIONS

- Data Science for Engineers – IIT Madras
- Natural Language Processing – IIT Kharagpur
- Data Analysis with Python – FreeCodeCamp
- Social Network Analysis – IIT Ropar
- Prompt Engineering with OpenAI – Columbia University
- Generative AI Masterclass – Udemy